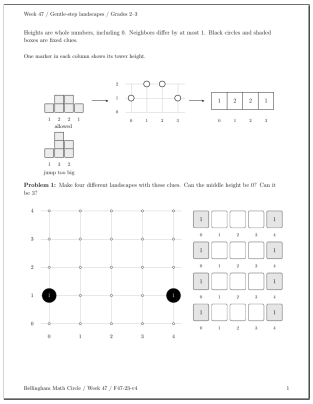
Week 47 / Gentle-step landscapes / Problem menu

for adults: pick, then print


Week 47 / Gentle-step landscapes / Grades 2–3

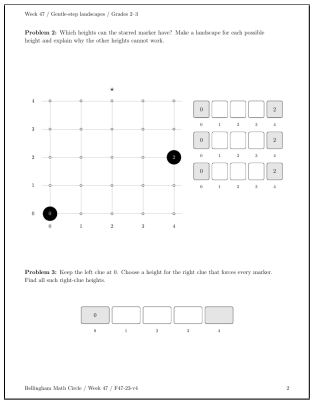
Heights are whole numbers, including 0. Neighbors differ by at most 1. Black circles and shaded boxes are fixed clues.

One marker in each column shows its tower height.

Problem 1: Make four different landscapes with these clues. Can the middle height be 0? Can it be 3?

Bellingham Math Circle / Week 47 / F47-23-v4 1

Different landscapes between two clues of height 1
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route
wording differs by band

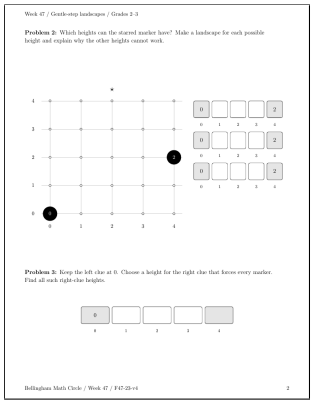

Week 47 / Gentle-step landscapes / Grades 2–3

Problem 2: Which heights can the starred marker have? Make a landscape for each possible height and explain why the other heights cannot work.

Problem 3: Keep the left clue at 0. Choose a height for the right clue that forces every marker. Find all such right-clue heights.

Bellingham Math Circle / Week 47 / F47-23-v4 2

How high and how low can the starred marker go?
K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2
first route
wording differs by band

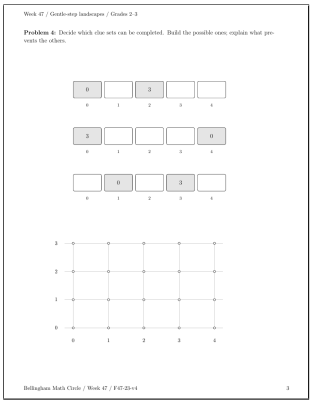

Week 47 / Gentle-step landscapes / Grades 2–3

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Bellingham Math Circle / Week 47 / F47-23-v4 2

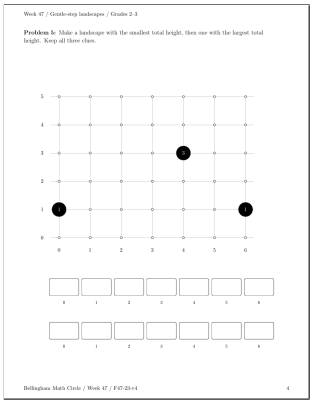
A right clue that forces every marker
K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3
first route
wording differs by band


Week 47 / Gentle-step landscapes / Grades 2–3

Problem 4: Decide which clue sets can be completed. Build the possible ones; explain what prevents the others.

Bellingham Math Circle / Week 47 / F47-23-v4 3

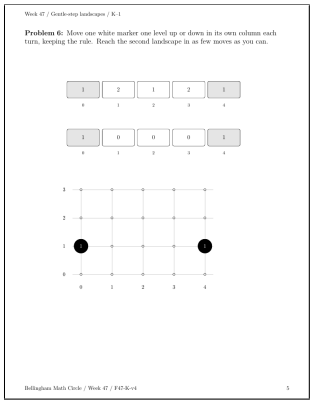
Which clue sets can be completed?
K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4
return route
wording differs by band


Week 47 / Gentle-step landscapes / Grades 2–3

Problem 5: Make a landscape with the smallest total height, then one with the largest total height. Keep all three clues.

Bellingham Math Circle / Week 47 / F47-23-v4 4

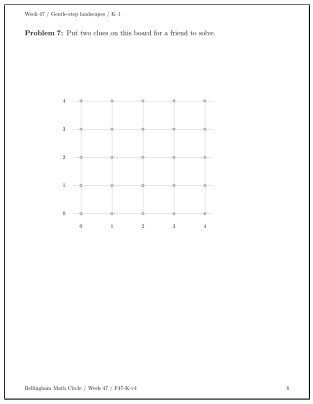
Lowest and highest landscapes on seven sites
K–1 P5 · Grades 2–3 P5 · Grades 4–5 P5
return route
wording differs by band


Week 47 / Gentle-step landscapes / K–1

Problem 6: Move one white marker one level up or down in its own column each turn, keeping the rule. Reach the second landscape in as few moves as you can.

Bellingham Math Circle / Week 47 / F47-K-v4 5

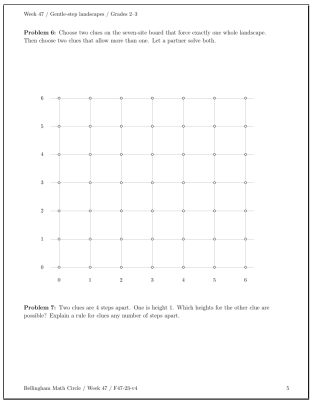
Fewest one-level moves between two landscapes
K–1 P6
return route


Week 47 / Gentle-step landscapes / K–1

Problem 7: Put two clues on this board for a friend to solve.

Bellingham Math Circle / Week 47 / F47-K-v4 6

Invent two clues for a friend
K–1 P7
return route

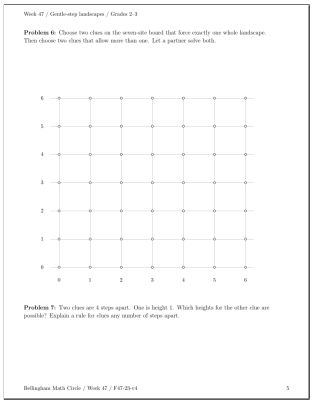

Week 47 / Gentle-step landscapes / Grades 2–3

Problem 6: Choose two clues on the seven-site board that force exactly one whole landscape. Then choose two clues that allow more than one. Let a partner solve both.

Problem 7: Two clues are 4 steps apart. One is height 1. Which heights for the other clue are possible? Explain a rule for clues any number of steps apart.

Bellingham Math Circle / Week 47 / F47-23-v4 5

Two clues that force one landscape, and two that do not
Grades 2–3 P6
return route

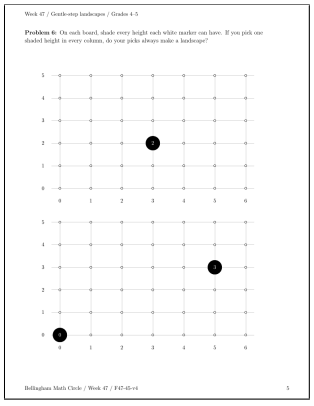

Week 47 / Gentle-step landscapes / Grades 2–3

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Bellingham Math Circle / Week 47 / F47-23-v4 5

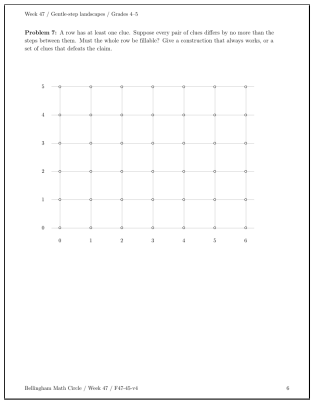
Possible heights for a clue four steps away
Grades 2–3 P7
return route


Week 47 / Gentle-step landscapes / Grades 4–5

Problem 6: On each board, shade every height each white marker can have. If you pick one shaded height in every column, do your picks always make a landscape?

Bellingham Math Circle / Week 47 / F47-45-v4 5

Allowed heights around one clue and around two
Grades 4–5 P6
return route


Week 47 / Gentle-step landscapes / Grades 4–5

Problem 7: A row has at least one clue. Suppose every pair of clues differs by no more than the steps between them. Must the whole row be fillable? Give a construction that always works, or a set of clues that defeats the claim.

Bellingham Math Circle / Week 47 / F47-45-v4 6

Must pairwise-consistent clues be fillable?
Grades 4–5 P7
return route